

Survivor Analysis: The Impact of Advantages

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Purpose

The purpose of this project is to analyze the relationship between the advantages that castaways obtain and their final standing in the competition.

Context

48 seasons of the United States version of Survivor have aired as of the completion of this analysis. Advantages were introduced in season 11, so there have been 38 seasons where advantages may have impacted a castaway's final standing.

In Survivor, someone's final standing can fall into four categories: Voted Out Pre-Jury, Jury, Finalist, and Winner. Partway through the season, the tribes merge. Anyone who was voted out before that time does not become a member of the jury. Each subsequent player who is voted out joins the jury until they reach the point of the final two or final three (depending on the season). Of the individuals who reach the final tribal council, one becomes The Sole Survivor, aka the winner.

Survivor is generally considered to have "eras" in which the production of the show and style of gameplay have themes that lend them to be grouped together and distinguished from other eras. Because advantages did not exist within Survivor when it debuted, and the ways in which advantages have been incorporated into gameplay have evolved over the years, I am going to categorize each season into an era.

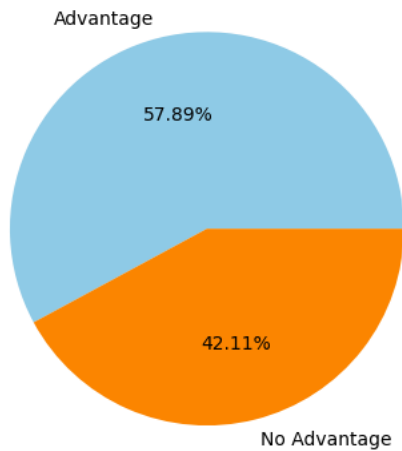
There is no official categorization of these eras. For the purposes of this project, seasons are grouped in the following manner based on a rough consensus among fans:

- Old School (pre-advantage):1-10
- Old School (with advantages): 11-20
- New School: 21-40
- New Era: 41 to present

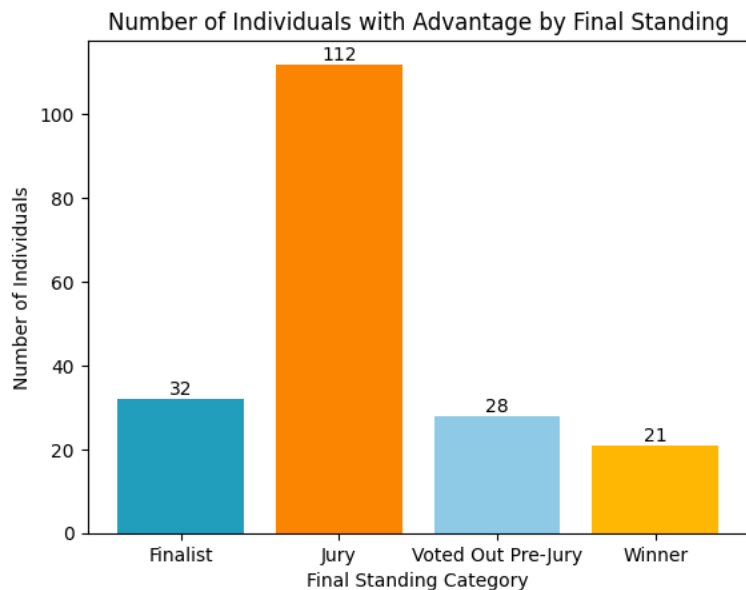
Key Findings

There are 38 winners in seasons with advantages. 57.89% of winners had an advantage, while 42.11% did not.

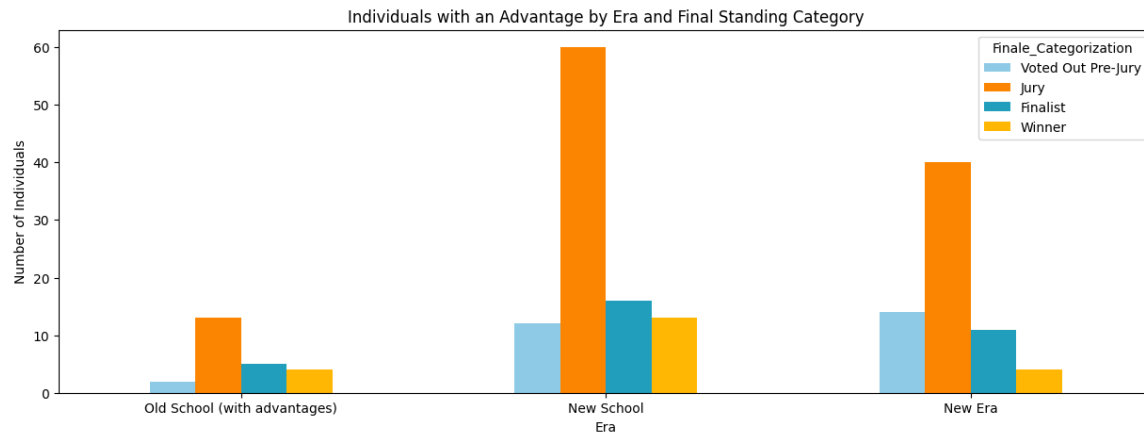
Percentage of Winners With or Without an Advantage



193 castaways have had advantages, with approximately 11% of those individuals becoming The Sole Survivor. 85% of individuals with advantages have made it to the jury or beyond, suggesting that having an advantage helps individuals get to the merge.



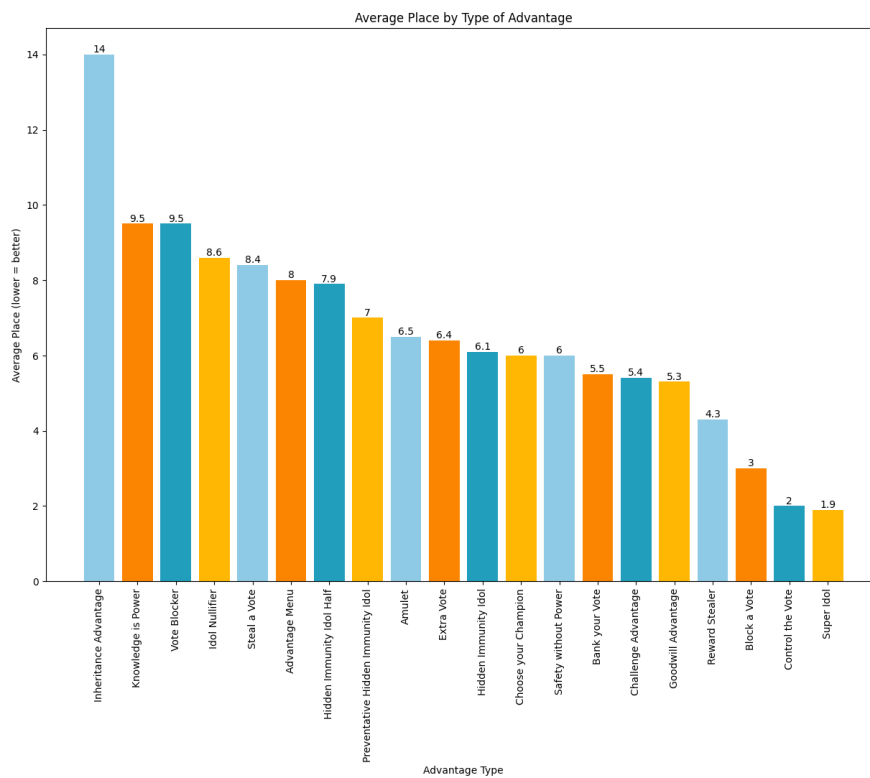
The correlation between having an advantage and a castaway's final standing is fairly consistent across seasons, but a higher percentage of individuals have been voted out pre-jury in the New Era. In the Old School (with advantages) and New School era, the number of people voted out pre-jury was lower than the other categories, but in the New Era, more people have been voted out pre-jury than those who have been Finalists or Winners.



In addition to “Finale Categorization”, “Place” can also be used as a metric for a castaway’s longevity in the competition. Place is the finishing order of the season, with one signifying the individual who won and the numbers increasing by one until reaching the first player to exit each season. Castaways with the Super Idol had the longest average duration in the game with an average Place of 1.9.

The Inheritance Advantage is the only type with a double-digit average finishing place.

Hidden Immunity Idols, which are by far the most common type of advantage, will, on average, get a player to 6th place.



There are 20 types of advantages, but individuals who have won have only had 6 of the types. The percentage of individuals with each type of advantage who have won varies. For instance, 100% of the individuals who have had the Block a Vote advantage have won, whereas only 18% of individuals with an Extra Vote have won. This may suggest that some types of advantages are more helpful for securing a win than others.

Advantage Type	Percentage of Players with Advantage Type Who Have Won
Block a Vote	100%
Super Idol	80%
Reward Stealer	50%
Hidden Immunity Idol	37%
Steal a Vote	25%
Extra Vote	18%

Limitations

There are a number of limitations to this analysis in its current form. Firstly, the number of finalists has changed throughout the seasons. Originally, there was a winner and one runner-up, but more recently, there has been a winner and two runners-up. Additionally, the number of castaways per season and the number of individuals on the jury have varied. This means that in some seasons there are more pre-jury boots than others.

Another limitation is that each type of advantage does not appear with the same frequency. Hidden immunity idols are much more common than any other advantage. This lack of consistency in how many times each type of advantage has appeared in the show impacts findings such as the average place by advantage type.

A limitation of how the data was cleaned for this project is the failure to account for medical evacuations when determining “finale categorization”. Due to not cross-referencing a castaway’s “Result” with their designation of Jury, Finalist, and Winner in the original data source, it is possible that individuals who were medically evacuated post-merge are incorrectly categorized as “Voted Out Pre-Jury” due to their lack of participation as a jury member in the final tribal council.

Conclusion

Advantages seem to help castaways remain in the competition longer, but the impact varies by type of advantage and the era in which the season occurs. There is a positive correlation between having an advantage and winning the competition.